Last Updated: October 2026

YUSUF OLANIYI

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SUMMARY

I am an AI-focused engineering graduate with experience building AI solutions for structural damage detection and modelling a digital twin system. I am skilled in Python, computer vision, data analysis, and machine learning. I am interested in conducting research that promotes safe engineering practices using emerging technology like AI and Digital Twins.

EDUCATION

Carnegie Mellon University Aug. 2025 - Dec.2026
MSc., Engineering Artificial Intelligence
Academic Standing: 3.88/4.00
Selected Coursework: Machine Learning, Deep Learning, and Digital Twins.

University of Lagos Nov. 2024
Bachelor of Science, Civil Engineering
Academic Standing: 4.73/5.00 (First-Class) (**Class Rank:** Top 1% - 1/76).

RESEARCH EXPERIENCE

Nanyang Technological University Singapore
Research Intern Feb 2026 - Mar 2026
Title: Multi-Region Corrosion Detection and Morphological Characterization Using Computer Vision (Poster)
Supervisor: Asst. Prof. Fu Yuguang (yuguang.fu@ntu.edu.sg).

- Conducted a literature review to identify gaps and formulate a methodology for vision-based corrosion severity estimation.
- Developed and evaluated deep learning segmentation models (SegNet, SegFormer), achieving up to 89% F1-score.
- Implemented post-processing (connected components, contour analysis) to extract region-level geometric features.

ML-Collective Lagos, Nigeria
Lead Investigator May 2024 - Nov 2025
Title: AfriCrack Benchmarking Domain-Shift Robustness of Crack Segmentation Models in African Infrastructure (Preprint)
Collaborators: Temitope Ibrahim, Zaynab Awofeso, and Ifihan Olusheye.

- Led the research pipeline (literature review, data collection, and documentation) to develop 1,000 images of cracks from Nigerian buildings, annotated using Roboflow to promote diversity in the images used for crack detection.
- Trained and evaluated CNN-based models and Vision Transformer (SegFormer) using PyTorch, revealing a ~50% IoU performance drop for models trained on controlled-condition datasets (SDNET2018) when transferred to AfriCrack.
- Evaluating the impact of using Magnitude-Aware Linear Attention as SegFormer's attention mechanism on computational efficiency and accuracy.

Dept. of Civil and Environmental Engr, University of Lagos Lagos, Nigeria
Graduating Project Thesis Aug. 2024
Title: Determination of the Natural Frequency and Mode Shape of the Beam Using Analytical and Finite Element Methods with Python. (GitHub Repository)
Supervisor: Dr. A.O Oyelade (aoyelade@unilag.edu.ng)

- Implemented a Python function to calculate the deflection of an Euler-Bernoulli beam using Rayleigh-Ritz and the principle of virtual work.

- Wrote a Python function to calculate the natural frequency of beams with three end conditions (clamp-clamp, clamp-simply supported, and simply supported at both ends).
- Wrote a Python function to calculate the natural frequency of tapered beams with two end conditions (clamp-clamp and clamp-simply supported).

PUBLICATIONS AND PRESENTATIONS

Yusuf, O., Olusheye, I., Awofeso, Z., & Amosa, T. I. (2026). AfriCrack: A Pixel-Level Crack Segmentation Dataset for Domain-Shift Research in Sub-Saharan Africa [Dataset]. Zenodo. https://doi.org/10.5281/zenodo.23735697

Global Connect Fellowship, 2026 - Multi-Region Corrosion Detection and Morphological Characterization Using Computer Vision. (Poster)

Raji, M.A., Falola, B.M., **Olaniyi, Y.A.** et al. (2024) Prediction of compressive strength of nano silica and micro silica from rice husk ash using multivariate regression models." *AI Civ. Eng. 3, 22* . https://doi.org/10.1007/s43503-024-00043-5

Deep Learning IndabaX Nigeria, 2024 - Crack detection in a building using a convolutional neural network. (Poster)

Big 5 Construct Nigeria, 2024 - Tech, Innovation, and Startups in AEC: Where Are We as an Industry? (Speaker Certificate)

RELEVANT PROJECTS

Railway Crossing Digital Twin System (GitHub Repo) Apr 2026

- Created the framework for a digital twin-based railway pedestrian crossing system.
- Implemented train and signal twin models with state estimation, uncertainty quantification, and confidence-aware decision-making.
- Built a real-time communication and visualization pipeline using Python, MQTT (Mosquitto), and Streamlit for live system monitoring.

PROFESSIONAL EXPERIENCE

Carnegie Mellon University, Africa Kigali, Rwanda
Graduate Teaching Assistant (Application of AI in Africa) Sep 2026 - Dec 2026

- Provided feedback to 79 students on both individual writing and group presentations.
- Contributed to course development by providing feedback based on my experience and observations from students' submissions.

Edubrige Academy Lagos Lagos, Nigeria
SQL Facilitator Aug 2023 - Nov 2024

- Designed and delivered a full SQL training curriculum for early-career professionals, covering querying, data cleaning, joins, subqueries, and real-world analytical workflows.
- Developed hands-on training materials, exercises, and capstone projects that helped learners apply SQL skills to practical business and data scenarios.
- Taught and mentored 40+ early-career professionals, improving their analytical capabilities and confidence in using SQL for data-driven decision-making.

Ove Arup and Partners Lagos, Nigeria
Civil Engineer Intern Jan. 2023 – Jun. 2023

- Conducted NDTs (Rebound Hammer, Ultrasonic Pulse Velocity, and Torque Wrench) on a 20,000 sqm factory facility and three residential apartments.
- Documented the results of the NDTs conducted on the factory facility, which helped the client make a data-driven decision in the acquisition of the facility.
- Remodelled rebound hammer results Excel template for non-technical personnel use.

TechPet Global Rivers, Nigeria
Machine Learning Engineer Intern Jun 2022 – Oct 2022

- Collaborated with a cross-functional team to develop a K-means clustering algorithm, classifying 100,000+ oil wells to help the client classify oil wells while making decisions.
- Developed an ML model to detect anomalies in heavy equipment using the PyCaret Library with a team of two within 24 hours of learning to prevent operational damage.
- Analysed 100,000+ sales data records with Python to derive insights on factors contributing to sales to help clients make data-driven decisions.

SKILLS AND OTHERS

Engineering: AutoCAD, Revit, Tekla Structures.
Research Tools: LaTeX, Image.io, Web of Science, Zotero.
Programming and Query Languages: Python and SQL.
AI & Machine Learning: TensorFlow, PyTorch, OpenCV, Scikit-Learn, Pandas, NumPy.
Data & Analytics: Power BI, Tableau, Microsoft Excel, Google Workspace
Soft Skills: Critical thinking, Problem Solving, Communication, Collaboration.
Selected Certifications: Deep Learning Specialization (deeplearning.ai), Machine Learning with Python (Coursera), Data Analyst Nanodegree (Udacity), Statistics for Data Science (Coursera).
Languages: English (fluent, spoken and written), Arabic (Beginner in reading & writing)
Interests: Education, Artificial Intelligence, and Engineering.

LEADERSHIP EXPERIENCE

Engineering Muslim Students' Society of Nigeria Lagos, Nigeria
Program Coordinator Jul 2023 – Aug 2024

- Increased the average attendance for weekly activities by 71% by introducing new activities to the program.
- Led a team of 9 that raised 1.2 million naira (~$760) in a month for the annual event.
- Managed a team of 10 that organised an event on Industry 5.0 for 250+ students.
- Initiated a community for females to drive cooperation and collaboration among them.

Nasir Giwa Mentorship Platform Lagos, Nigeria
Lead, NGM Academic Excellence and Scholarship Support Access Jan 2023 – Dec 2024

- Supported two NGM members through the application process for the NHEF scholarship, which helped them get 3.2 million Naira (~$4100) worth of training.
- Pioneered the state-wide annual essay competition for secondary school students by guiding the team in planning and executing the competition.
- Organized two sessions for 60 high schoolers on mentorship and academic excellence.

HONORS, AWARDS, AND RECOGNITIONS

- **Nanyang Technological University -** Global Connect Fellowship (Top 1.5%) 2026
- **University of Lagos**- Best Graduating Student (Civil Engineering) 2025
- **Deep Learning IndabaX Nigeria** - First Runner-Up Poster Presenter 2024
- **Nigeria Higher Education Foundation Scholars Program** - Scholar (Top 10%) 2023
- **Shell AI EV Hackathon -** Top 14% (Global Rank) 2022
- **The Bridge Program** - Fellow (Top 5%) 2022
- **MSSN UNILAG Tutorial** - Tutor of the Year (Top 1 of 25) 2021